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Lattice Boltzmann Simulation and Experimental Verification of In-Situ Gas Density and Storage in Shale Rocks

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Abstract

This study integrates pore-scale lattice Boltzmann (LB) simulations with micro-CT imaging to investigate gas densification and enhanced storage in a Haynesville shale sample, with a focus on fluid-wall interactions under nanoscale confinement. Xenon gas invasion experiments combined with X-ray imaging were conducted to directly measure in-situ gas density distributions. An averaged gas density 2.6 times the density of free-space xenon under the same pressure and temperature conditions was observed. A single-component pseudo-potential LB model was developed, incorporating adsorption effects via wall-induced attractive forces characterized by a tunable interaction strength. Haynesville sample pore structure was reconstructed using the Quartet Structure Generation Set (QSGS) method, with pore size distributions (PSD) derived from mercury intrusion porosimetry (MIP) experiments. By calibrating the simulation results against the in-situ xenon density obtained from micro-CT analysis, the strength of fluid-wall interactions and the extent of adsorption were quantitatively determined. It was observed that fluid-wall interactions significantly influence the density of the adsorbed-layer and transition layer. In small pores, the adsorbed-layer occupies a larger volume fraction, amplifying the influence of fluid-wall interactions on the overall densification ratio and storage capacity. As pore size increases, the impact of these interactions on densification and storage enhancement diminishes. Simulations using the MIP-derived PSD accurately reproduced the observed gas density distributions. It is observed that PSD is the primary factor controlling gas densification and apparent storage capacity, as smaller pores and narrow throats allow the adsorbed layer to occupy a large fraction of the available volume, markedly increasing in-situ fluid density. In contrast, structural features such as grain size and coordination number have little impact under equilibrium conditions.

Background

Compared with conventional reservoirs, shale gas reservoirs exhibit two fundamental differences (Zhao et al., 2016). First, most pores in shale are at the nanometer scale, significantly smaller than those in conventional reservoirs (Chen et al., 2021; Liu et al., 2018). Second, the presence of organic matter in shale introduces strong fluid-wall interactions due to the adsorption affinity between the organic matrix and gas molecules (Babatunde et al., 2022). The primary mechanism responsible for fluid densification is the adsorption of fluids by the organic matrix in shale, which significantly increases the near-wall fluid density. Moreover, the extremely small pore size amplifies the volumetric contribution of the adsorbed layer, thereby elevating the apparent fluid density at the macroscale.

The phenomenon of abnormally elevated fluid density in shale reservoirs has been experimentally observed (Chakraborty et al., 2020; Chakraborty & Lou et al., 2022; Lou et al., 2022). In these studies, X-ray microtomography (X-ray micro-CT) imaging was employed to quantify the gas storage capacity of shales from different sources. X-ray CT is a non-destructive imaging technique that enables the acquisition of three-dimensional (3D) structural information of rock cores. In a CT system, X-rays emitted from the source pass through the specimen and are recorded by a detector to generate two-dimensional (2D) projection images. Subsequently, by collecting multiple X-ray measurements along different paths, a 3D reconstruction of the sample can be obtained, allowing direct visualization of its internal structure (Zhang et al., 2021). The reconstructed 3D images represent maps of X-ray attenuation values within a core sample, which can be correlated with the density of the fluid present in the pore space by using image subtractions of initial (fresh) and gas saturated (equilibrium) images. Consequently, X-ray micro-CT imaging can be used to determine the in-situ fluid density within shale core samples (Chakraborty et al., 2020). Experimental results have demonstrated that, under equilibrium PVT conditions, the in-situ fluid density in various shale cores is consistently higher than the theoretical value, a phenomenon referred to as fluid densification.

As a mesoscopic approach that is not confined to the continuum hypothesis, the Lattice Boltzmann Method (LBM) can simulate fluid behaviors in larger flow spaces without excessive computational resource consumption. Meanwhile, it allows for the flexible incorporation of fluid-wall interactions, thereby enabling the simulation of fluid adsorption in organic porous media under various experimental conditions. To date, a series of advancements have been achieved in LBM-based adsorption simulations (Xu et al., 2019; Xu et al., 2020; Zhang et al., 2023; Guo et al., 2024). In LBM-based adsorption simulations, determining the strength of fluid-wall interactions has emerged as a critical issue to be addressed. In numerous studies, the fluid-wall interaction strength has been treated as a variable (Guo et al., 2016; Huang et al., 2021), with the influence of adsorption strength on fluid behaviors investigated by adjusting the adsorption intensity of wall surfaces on fluid particles. Furthermore, matching the results of molecular dynamics simulations with LB simulations represents another feasible approach to determine the fluid-wall interaction strength (Wang et al., 2022, 2024a, 2024b).

In the present study, we integrate LB simulations with the gas injection experiments followed by X-ray micro-CT image analysis. Specifically, the in-situ fluid density obtained by X-ray micro-CT imaging is used to directly determine the strength of fluid-wall interactions, further investigating how pore size, porous media structure, particle size, and coordination number distribution affect the overall fluid densification ratio. LB-based simulations are performed to resolve fluid density at pore-scale, allowing for localized comparisons among the adsorbed layer, transition region, and bulk fluid. By combining experimental imaging dataset with pore-scale simulations, this work provides new insights into fluid behavior within realistic shale nanopore environments.

Lattice Boltzmann Method for Enhanced Gas Storage Modeling in Porous Media

The Lattice Boltzmann model employed in this study is formulated based on the single-relaxation-time (SRT) Lattice Boltzmann equation. Within this framework, particle dynamics in physical space are decomposed into two fundamental processes: collision and streaming. To accurately incorporate external forces, the Exact-Difference-Method (EDM) scheme (Kupershtokh et al., 2009) was employed. The governing evolution equation of the model is given by Equation 1

$$f_\alpha(\boldsymbol{r}+\boldsymbol{e}_\alpha\delta_t, t+\delta_t) - f_\alpha(\boldsymbol{r}, t) = -\tfrac{1}{\tau}\left[f_\alpha - f_\alpha^{\mathrm{eq}}(\rho, \boldsymbol{u})\right] + f_\alpha^{\mathrm{eq}}(\rho, \boldsymbol{u}+\Delta\boldsymbol{u}) - f_\alpha^{\mathrm{eq}}(\rho, \boldsymbol{u})$$
$$f_\alpha^{eq} = \rho w_\alpha\left[1+\frac{\boldsymbol{e}_\alpha\boldsymbol{u}}{c_s^2}+\frac{(\boldsymbol{e}_\alpha\boldsymbol{u})^2}{2c_s^4}-\frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right] \quad \text{Eq. 1}$$
$$\Delta\boldsymbol{u} = \frac{\boldsymbol{F}\delta_t}{\rho}$$

f_α denotes the distribution function in discrete velocity space along direction α, while f_α^{eq} represents its equilibrium distribution function, **r** is the spatial position of the particle, $\mathbf{e}_\alpha$ is the velocity in α direction, t is time, δ_t is the time step, and τ is the relaxation time, ρ is fluid density, $\boldsymbol{u}$ is macroscopic velocity of fluid, c_s is the lattice sound speed and equal to $1/\sqrt{3}$, and w_α is the weight coefficient. $\boldsymbol{F}$ denotes the resultant external force acting on fluid component. The model requires the specification of two distinct types of interaction forces: the interparticle force $\boldsymbol{F}(\boldsymbol{x})_{\mathrm{in}}$ (Kupershtokh et al., 2009) between fluid particles and the interfacial force $\boldsymbol{F}(x)_{\mathrm{out}}$ (Li et al., 2014) between the wall surface and fluid particles. For the mathematical formulation of these interaction forces, the computational scheme proposed by Huang et al. (2021) is implemented

$$\boldsymbol{F}(\boldsymbol{x})_{\mathrm{in}} = -\beta c_0 g\psi(\boldsymbol{x})\nabla\psi(\boldsymbol{x}) - \frac{1-\beta}{2}c_0 g\nabla\left(\psi^2(\boldsymbol{x})\right) \quad \text{Eq. 2}$$

$$\boldsymbol{F}(\boldsymbol{x})_{\mathrm{out}} = -g_w\psi(\boldsymbol{x})\sum\omega_i\psi(\boldsymbol{x})s(\boldsymbol{x}+\boldsymbol{c}_i) \quad \text{Eq. 3}$$

$$\boldsymbol{F} = \boldsymbol{F}(\boldsymbol{x})_{\mathrm{in}} + \boldsymbol{F}(\boldsymbol{x})_{\mathrm{out}} \quad \text{Eq. 4}$$

β is a tunable parameter, c_0 is a constant with a value of 6, g is a parameter that controls the interaction between fluid particles, $\psi(\boldsymbol{x})$
is the pseudo-potential, g_w is a parameter that controls the interaction between the wall surface and the fluid particles, ω_i is the weighting parameter, s ($\boldsymbol{x}$ + $\boldsymbol{c}_i$) equals 1 when $\boldsymbol{x}$ + $\boldsymbol{c}_i$ points to a solid node and 0 when points to a fluid node. In this LB model, the thermodynamic behavior of the fluid is governed by the Peng-Robinson equation of state, which provides a fundamental description of the pressure-density-temperature relationship. The accurate characterization of fluid pressure through this equation is essential for determining the pseudo-potential in the computational framework. The mathematical formulations of both the Peng-Robinson equation of state and the corresponding pseudo-potential are explicitly given in Eqs. 5 and 6, respectively.

$$p_{\mathrm{EOS}} = \frac{\rho RT}{1-b\rho} - \frac{a\alpha(T)\rho^2}{1+2b\rho-b^2\rho^2}$$
$$a = 0.45724\frac{R^2T_c^2}{P_c} \quad b = 0.0778\frac{RT_c}{P_c} \quad \text{Eq. 5}$$

$$\psi = \sqrt{\frac{2\left(p_{EOS}-c_s^2\rho\right)}{c_0 g}} \quad \text{Eq. 6}$$

ρ is fluid density, R is gas constant, T is temperature, a and b are state equation parameters, T_c and P_c are the critical temperature and critical pressure of fluid component. In the lattice Boltzmann model incorporating the equation of state, the EOS parameters are determined following the methodology established by Yuan et al. (2006). Specifically, the coefficients are set as a = 2/49, b = 2/21, and R = 1. Notably, the interaction parameter g, which governs fluid-fluid interactions, does not influence the magnitude of interparticle interaction forces. The sole requirement is to maintain a positive value for the square root term in Eq. 6.

Experimental Setup and Results

The experimental results used in this study are from the work of Lou et al.(2022) and briefly summarized here. Gas invasion experiments coupled with micro-CT imaging were conducted using the setup described in Chakraborty et al. (2020) and Lou et al.(2022), as shown in Fig. 1. Xenon gas (purity >99.5%) was used as a high-contrast probe. The tested Haynesville core plug measured 12 mm in diameter and 35.7 mm in length, with a porosity of 9.03%, TOC of 5.6% and permeability ranging from 100 to 500 nD. Xenon gas was injected as a single pulse at approximately 400 psi, after which the ball valve was closed, creating a closed system. Gas from the annular space gradually entered the sample's pore space until pressure equilibrium was established between the two, as indicated by a stabilized annular pressure profile over time.

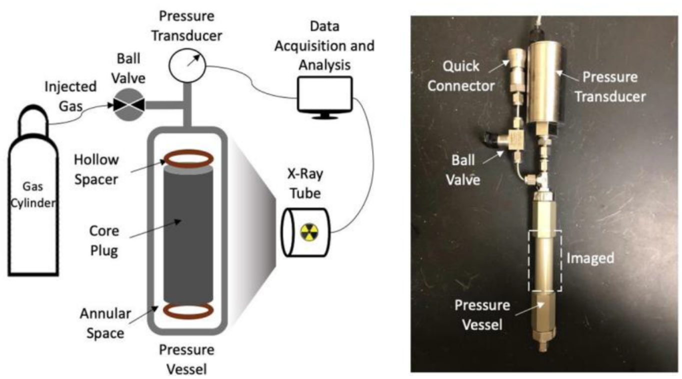

Figure 1—(left) Schematics of experimental setup for gas injection and pressure fall-off measurement;(right) the actual experimental setup

For the Haynesville sample, the final equilibrium pressure was 300 psi, and pressure stabilization occurred after approximately four days. A X-ray micro-CT image provide a 3D map of X-ray attenuation values based on material density; thus, changes in X-ray attenuation values between the initial state (pre-xenon injection) and the equilibrium state images denote variations in xenon density.

The detailed calculation procedure is described in Chakraborty et al. (2020). By mapping the change in X-ray attenuation values to xenon density, in-situ density distributions were obtained, as illustrated in Fig. 2. The average xenon density within the analyzed sub-volume, based on the 3D distribution shown in Fig. 3, was calculated to be 326.05 kg/m^3, corresponding to a densification ratio of 2.6 relative to the free-space xenon density (128 kg/m^3) calculated using the equation of state.

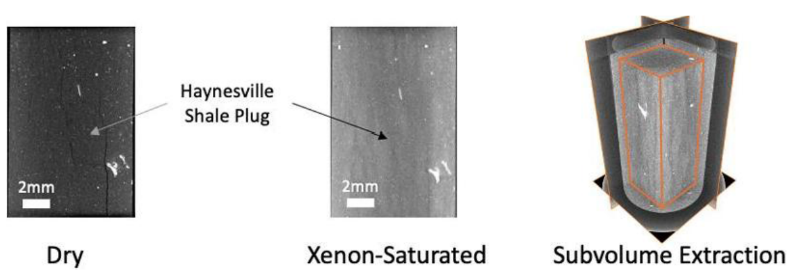

Figure 2—CT images of the Haynesville shale sample. Left to right: Two-dimensional longitudinal cross-sections of the shale sample at dry conditions (before xenon injection); same cross-section at equilibrium conditions; and extracted sub-volumes for analysis

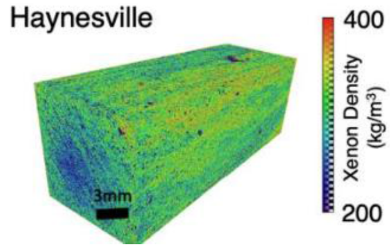

Figure 3—3D spatial density maps at equilibrium condition of the Haynesville shale sample

Gas Densification Simulation in Pore-Throat Geometries

For illustration, xenon gas invasion within the pore-throat channel shown in Fig. 4 was simulated using the proposed LB model with three interaction strengths (g_w = -2.1, -2.5 and -2.9), to examine how pore and throat size influence the gas density distribution in the porous space. The simulation was conducted at the same temperature as the experiment (526.77 °R). The model consists of three pores with diameters of 100 nm, 50 nm, and 80 nm, respectively, connected by throats of 21 nm, 9 nm, 3 nm, and 21 nm in diameter (from left to right). The outermost throats on both ends are open to the external environment, where boundary pressures are applied, matching the stabilized pressure in the annular space observed in the experiment (300 psia). Initially, the model was uniformly filled with xenon of free space density (128 kg/m^3) under ambient temperature and pressure. In this study, the simulated adsorption layer thickness was found to be approximately 1–2 lu. Considering that molecular simulations typically yield adsorption layer thicknesses in the range of 1.3–1.9 nm (Xu et al., 2024), setting the lattice spacing to 1 lu = 1 nm provides an appropriate representation of the adsorption layer thickness.

Figure 4—Schematic of a simple pore-throat structure

Once the density distribution within the pore-throat channel reaches equilibrium, the xenon density profiles in the pores and throats are analyzed, as shown in Figs. 5–7. Figure 5 displays the fluid density distribution in lattice Boltzmann units under a specified interaction strength parameter ($g_w = -2.5$). Figure 6 presents the corresponding cross-sectional density profiles within pores and throats of different diameters, converted into physical units for various values of g_w. The unit conversion is based on the simulation temperature and pressure, with the physical properties of xenon listed in Table 1. According to the conversion calculations, one dimensionless density unit corresponds to 15,3 14.16 kg/m^3. Further details of the conversion procedure are provided in Appendix A. As illustrated in Figs. 5 and 6, the attractive solid-fluid interactions induce a pronounced increase in fluid density adjacent to the walls, forming a high-density adsorption phase layer. Between this layer and the bulk fluid, a transition zone of finite thickness is observed, where the fluid density gradually decreases from the adsorption layer value to that of the bulk phase.

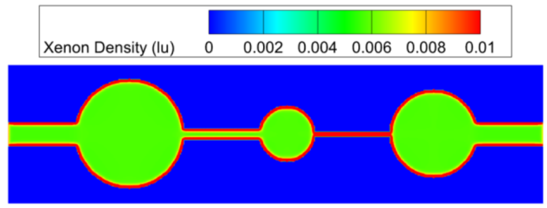

Figure 5—Equilibrium xenon density distribution in pore-throat channel

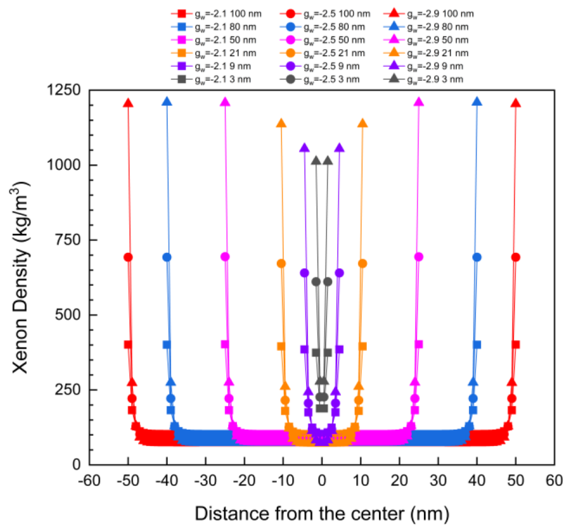

Figure 6—Distribution of xenon density in the vertical cross-section line in throat channels and pores of different diameters

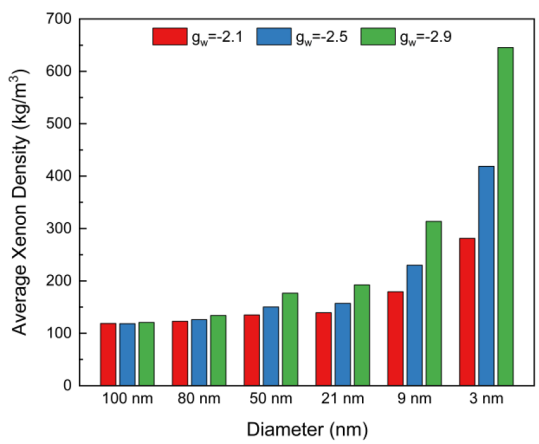

Figure 7—The average xenon density in pores/throats with different diameters under varying wall adsorption strengths

Table 1—Physical properties of Xenon

component	Xenon
critical temperature T_c, °R	521.53
critical pressure P_c, *psi*	847.31
acentric factor ω	0
Molecular weight *MW, lbmass/lbmole*	131.2

Take the case where $g_w = -2.5$ as an example, in the pore and throats with diameters higher than 20nm, it can be clearly observed that adsorption layer exhibits a density of approximately 690 kg/m³, while the bulk fluid density is around 90 kg/m³, slightly lower than the free-space xenon density. The thickness 1–2 nm and density of the adsorption layer remain essentially constant throughout the domain, a direct consequence of the fixed interaction strength parameter g_w. Such observation is consistent with previous findings based on molecular simulations (Xu et al., 2024; Deng et al., 2024; Zhang & Tang, 2024).

In the narrowest throat (3 nm), the adsorption layer density decreases slightly (to 610 kg/m3), whereas the bulk fluid density increases significantly (to 230 kg/m³). As the throat diameter increases to 9 nm and 21 nm, this effect diminishes: the bulk density converges toward 90 kg/m³, comparable to that in pores, while the adsorption layer density remains marginally lower. This phenomenon can be attributed to the overlap of transition zones in narrow throats (Hajianzadeh et al., 2023), which elevates the bulk fluid density, while competing wall forces from opposing surfaces reduce the adsorption layer density (Konstantakou et al., 2011). Furthermore, in smaller pores and narrower throats, the adsorption layer occupies a greater proportion of the total fluid volume, leading to a substantial increase in the average fluid density within the domain. This finding provides a mechanistic explanation for the densification phenomenon observed in the experiment.

Results and Discussion

To simulate conditions representative of the actual Haynesville shale sample, 2D porous models were reconstructed using the PSD curves experimentally obtained by Lou (2023). The best-fit log-normal distributions derived from MIP data are presented in Fig. 8. Note that the vertical axis labeled 'percentage' does not correspond to a true probability density function, but rather reflects an unformalized relative distribution. The fitted curve aligns well with the experimental data. The pore size distribution obtained from the MIP data is concentrated in the range of 3–16 nm, with an average pore diameter of 13.12 nm calculated from the fitted curve.

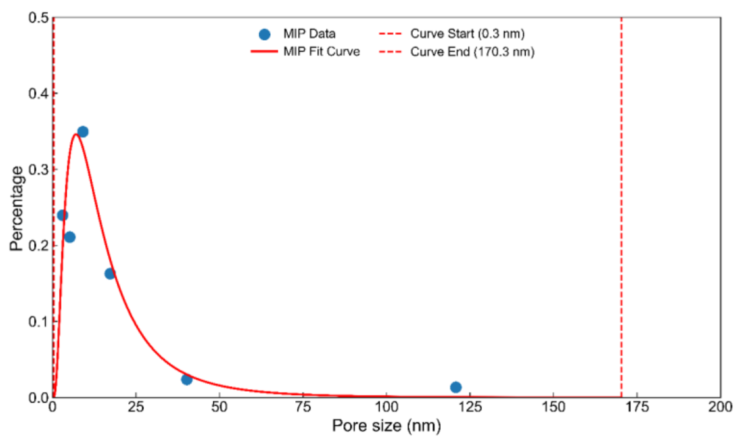

Figure 8—Pore throat distribution from Mercury intrusion porosimeter

Using the PSD obtained from the MIP best fit curve, we reconstructed a 2D porous model using the Quartet Structure Generation Set (QSGS) method (Wang & Pan, 2007). The porous model domain measures 500 nm by 500 nm, as shown in Fig. 9. White regions indicate the flow paths, including both pores and throats, while black regions correspond to the shale matrix. The lattice spacing was set to 1 nm, resulting in a grid of 500 by 500 lattice nodes.

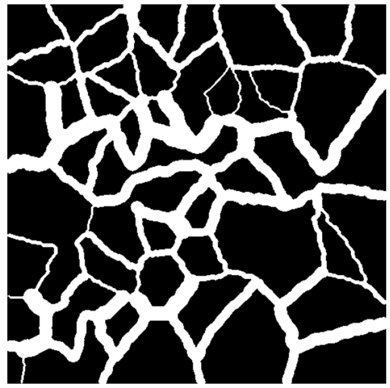

Figure 9—Porous model generated using MIP-derived PSD

To improve numerical stability, five layers of fluid nodes were added around the perimeter of the model. Constant-pressure boundary conditions were imposed along the edges of the domain, with the pressure set to the stabilized experimental value of 300 psi. The initial density throughout the domain was set to the free-space xenon density. Once the simulation began, xenon gas continuously entered the porous medium from the boundary regions, driven by both the attractive interactions between the solid walls and the fluid, as well as the density gradient between the boundary layer and the porous structure.

The apparent density of the porous domain is calculated as the average density over all lattice nodes located within the pore space, excluding those corresponding to solid regions and boundary layers. To identify the interaction strength that best reproduces the experimental results, a series of simulations were conducted under varying values of the wall-fluid interaction parameter g_w. By adjusting g_w and computing the resulting average Xenon density, it was found that $g_w = -2.43$ yields the best match, producing an apparent density of 326.47 kg/m^3, which closely agrees with the experimentally measured equilibrium value, as shown in Fig. 10. Figure 11 presents the domain-averaged fluid densities under different g_w values. As the strength of the fluid-wall interaction increases, the density of the adsorbed layer rises significantly, as shown in Fig. 6. Due to the relatively small pore sizes distribution in domain, this increase in adsorbed-layer density substantially impacts the overall fluid density. As a result, a clear increase is observed in domain-averaged fluid density with stronger interaction strength.

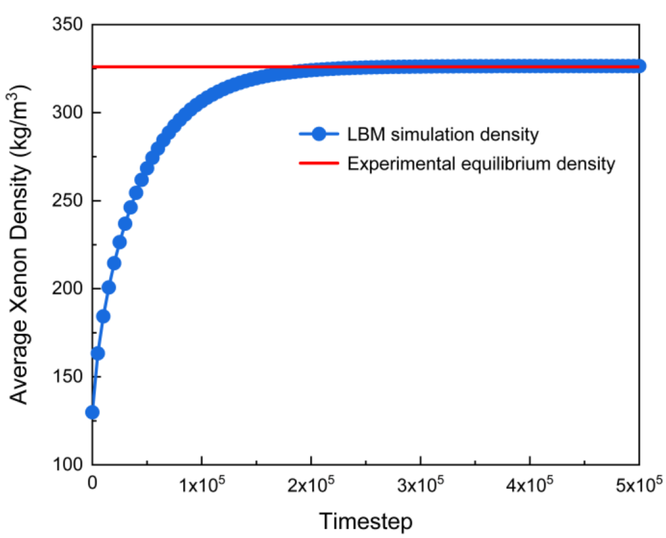

Figure 10—The comparison between experimental results and LBM simulation results

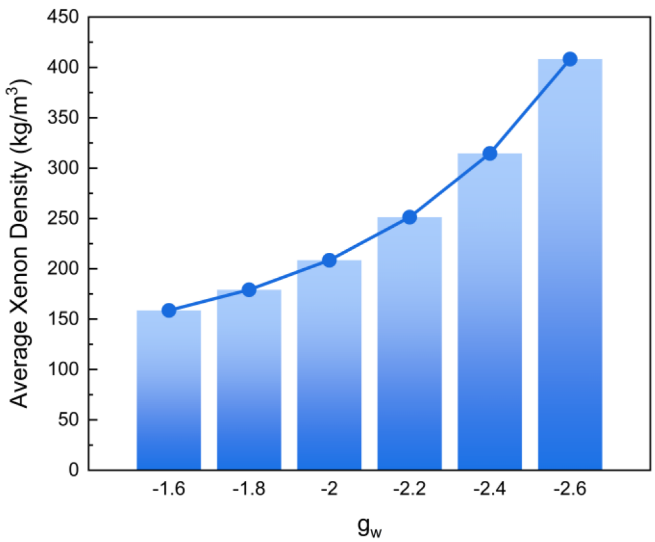

Figure 11—The LBM simulation results under different g_w values

Fig. 12 presents the evolution of the density distributions during the dynamic transport process, taken from initial (0 ts), intermediate (50000 ts) and equilibrium state (500000 ts). After the simulation begins, the density distribution shows a clear distinction between the adsorbed phase and the bulk fluid. In smaller pores and throats, the density increases sharply due to the dominant influence of the adsorbed phase. As xenon continues to enter the domain, both adsorbed and bulk phase densities increase, and propagation fronts are observed. Once the density field stabilizes, significant variations appear across pores and throats of different sizes.

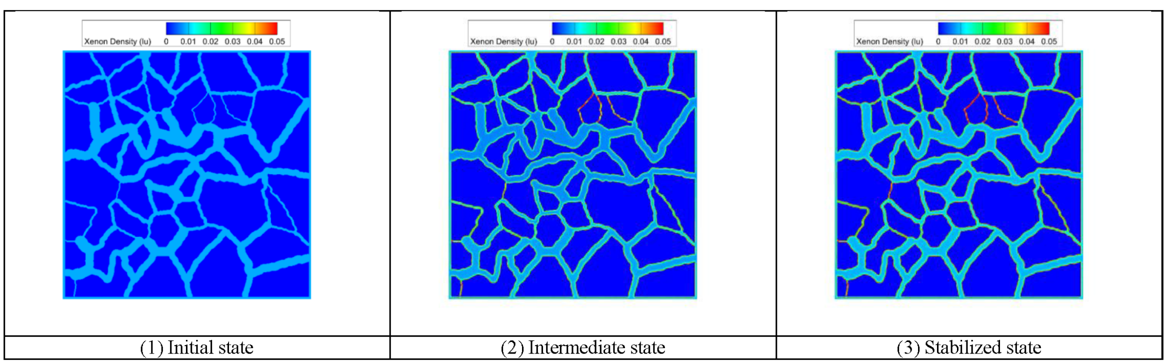

Figure 12—Wall adsorption causes densification of xenon in MIP-derived porous model

Sensitivity Analysis

Given the inherent randomness of the QSGS method used to generate the porous model, additional simulations were conducted to evaluate the sensitivity of the porous structure. Specifically, 10 additional simulations were performed under identical conditions to enable a sensitivity analysis. The Xenon density distributions obtained at equilibrium condition for these simulations are provided in Fig. 13. In the first 5 simulations, the pore size distribution, grain size distribution, and coordination number distribution closely resemble those of the reference pore network model shown in Fig. 9. In Case 6-10, the grain size and coordination number were intentionally modified to introduce greater structural variation. The resulting differences in domain-averaged xenon density, under the same interaction parameter $g_w = -2.43$, are summarized in Fig. 14.

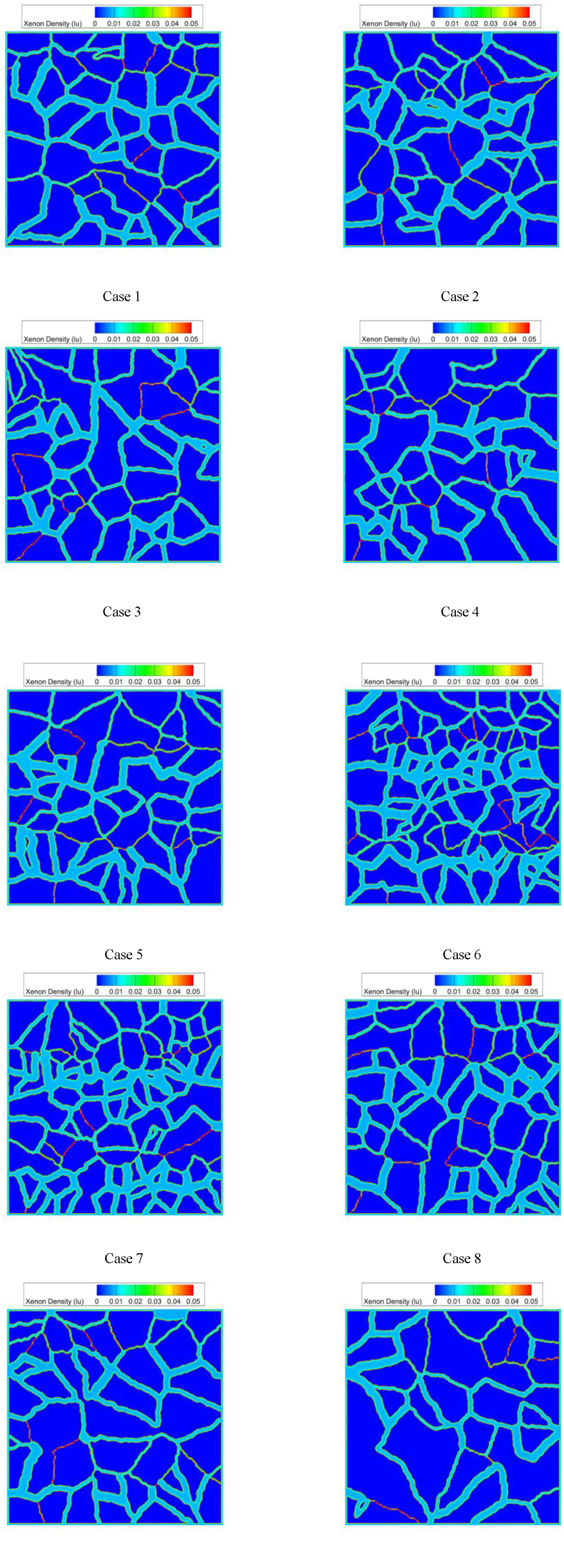

Figure 13—Equilibrium density distributions for all case tested. Case 6-10 are simulated core models with same PSD but different grain size and coordination number.

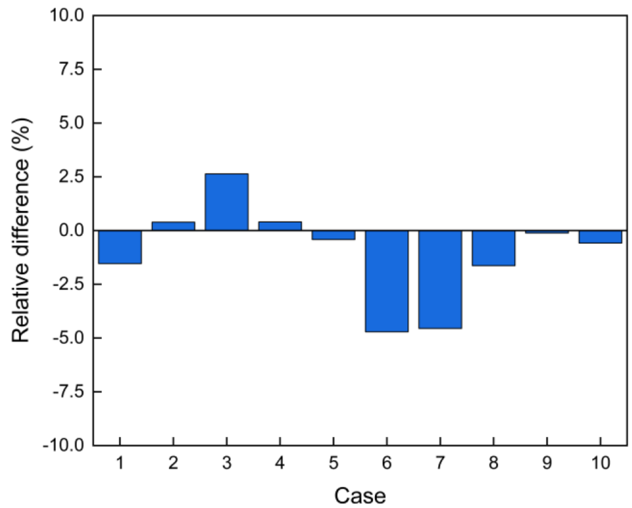

Figure 14—Equilibrium density comparisons. Case 6-10 are simulated on models with the same PSD but different grain size and coordination number.

As shown in Fig. 14, the stabilized average densities obtained from all ten simulation cases are in good agreement with the experimental value, indicating that the densification ratio under equilibrium conditions is largely independent of the specific geometry of the porous structure. Moreover, results from Case 6–10 show that variations in grain size and coordination number have only a minor influence on the domain averaged xenon density. Once the pore size distributions and wall-fluid interactions are defined, the gas storage capacity of the porous model appears to be effectively determined. However, it is important to note that the dynamic transport process may still be significantly affected by these structural parameters, which will be the focus of future investigation.

In this section, xenon adsorption simulations were conducted based on a 2D porous model reconstructed using the PSD obtained from MIP for the Haynesville shale sample. The experimentally-measured in-situ fluid density was successfully used to calibrate the wall-fluid interaction strength. In smaller pores, the adsorbed layer occupies a substantial portion of the pore volume, resulting in an average fluid density significantly higher than the free space density. However, as pore size increases, the volume fraction of the adsorbed layer decreases, leading to a reduced degree of densification. This finding was validated by simulations conducted in both simplified pore-throat channel mode and porous network models. Notably, the equivalent porous model was constructed using PSD derived from MIP. While a good match with experimental gas density was achieved by tuning the fluid-wall interaction parameter, this does not imply that the MIP-derived PSD is uniquely representative; alternative PSDs obtained from other experimental techniques (e.g. N_2 adsorption) may also yield satisfactory matches under different interaction strengths. Although uncertainties remain regarding the true pore size distribution, this study provides a practical approach for exploring gas densification in shale nanopores. The ability to reproduce experimental gas density by adjusting the fluid-wall interaction parameter highlights the key roles of pore size and fluid-wall interaction. In addition, simulations show that variations in grain size and coordination number have limited impact on the equilibrium average xenon density, indicating that gas storage capacity is mainly governed by the pore size distribution and wall-fluid interaction. These findings help clarify the dominant factors controlling adsorption-induced densification in nanoporous shales.

Conclusion

This study marks the first attempt to investigate adsorption-induced gas densification and enhanced storage in shale nanopores by integrating pore-scale simulations with experimental data. The equivalent porous model was constructed based on experimentally measured pore size distributions, and the simulated gas density was validated against micro-CT-based in-situ density measurements. A single component pseudopotential model of lattice Boltzmann method was developed, with adjustable fluid-wall interaction strength parameter to capture adsorption-driven densification in nanopores. By tuning the wall-fluid interaction parameter, we systematically adjust adsorption strength and compared the simulated xenon densities with experimentally measured in-situ value. The key findings are summarized as follows:

1. Wall-fluid interactions affect the densification ratio mainly by changing the density of the adsorbed layer and transition layer Their impact depends on pore size: in larger pores, the adsorbed layer occupies little volume and has limited effect on overall density, while in smaller pores, it takes up a larger portion and plays a greater role in increasing fluid density.
2. The pore size distribution plays an important role in determining gas densification and apparent storage capacity in nanoporous systems. In smaller pores and narrow throats, the adsorbed layer occupies a substantial portion of the available volume, resulting in higher in-situ average fluid densities. In extremely narrow throats and small pores, adsorbed fluid can occupy most of the available space, resulting in exceptionally elevated average densities.
3. While the PSD largely determine the apparent storage capacity, structural features such as grain size and coordination number have limited influence under equilibrium conditions.

This study shows that lattice Boltzmann simulations can reproduce the coexistence of adsorbed and bulk phases in shale nanopores, with good agreement to experimental density achieved by tuning wall-fluid interaction in a porous model based on MIP-derived PSD. However, PSD uncertainty remains a key limitation, as different experimental methods yield varying results. The current match does not imply uniqueness, and further work is needed to examine the impact of alternative PSDs and improve model representativeness. Additionally, this work focuses on equilibrium conditions, with a simulation setup slightly differing from that in Lou et al (2022) & Lou (2023). Future efforts should address transient gas transport and adsorption dynamics to better understand storage mechanisms in shale.

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Appendix A

To convert LB units to real units, Xenon was used as the component to calculate the conversion coefficients of the following 4 variables, as shown below:

$$a = 0.45724\frac{R^2T_c^2}{p_c}$$ Eq. A-1

$$b = 0.07780\frac{RT_c}{p_c}$$ Eq. A-2

Table A-1—Calculation of unit conversion coefficients

Parameter name	Real unit value	LB unit value	
Gas constant R	49712.69 lbm·ft²/(s² ·lbmol · R)	1	49712.69
PR-EOS parameter a	50659.015855 lbm/ lbmol · ft² · s² · °R	9/49	275838.62
PR-EOS parameter b	0.0130699 ft³/lbmol	2/21	0.13723395
molecular weight MW	131.2 lbmass/lbmole	1	131.2

Since the distance between adjacent nodes is 1nm (3.281×10^{-9}ft), therefore the conversion coefficients of length L, $\chi_L = 3.281 \times 10^{-9}$. Considering that the tension conversion coefficient can be calculated by the length conversion coefficient by Equation 8, the conversion coefficients of tension σ, $\chi_\sigma = 0.048054955$.

$$\chi_L = (\chi_\sigma)(\chi_b)^2(\chi_a)^{-1}$$ Eq. A-3

The conversion coefficient of mass M is calculated by Equation 9, $\chi_M = 3.37668823396 \times 10^{-23}$, so as to calculate the density transformation coefficient $\chi_\rho = 956.0316572$ by Equation 10. This indicates that 1 lu density = 956.03 lbm/ft³ = 15314.16 Kg/m³.

$$\chi_M = (\chi_{MW})(\chi_\sigma)^3(\chi_b)^5(\chi_a)^{-3}$$ Eq. A-4

$$\chi_\rho = \chi_M / \chi_L^3$$ Eq. A-5